

Roll no: 26k-0006

Ques no 5 (Smart Campus Parking and Access Management System)

Algorithm:

- 1) start
- 2) Initialize capacity
 - Max limits: ZoneA - Max = 20, ZoneB - Max = 40, ZoneC - Max = 15
 - Occupied Counters: ZoneA - Occ = 0, ZoneB - Occ = 0, ZoneC - Occ = 0
 - Statistics logs: Total - Processed = 0, Total - Accepted = 0, Total - Rejected = 0.
 - Vehicle subtype logs: Cars - Count = 0, Bikes - Count = 0, Vans - Count = 0.
- 3) Input no. of vehicles expected (Total - Expected)
- 4) WHILE (Total - Processed < Total - Expected):
 - a) Read inputs: Vehicle Type (C, B, V), User - Category (F, S, G), Permit - status (Y, N) and Emergency - status (Y, N)
 - b) IF Vehicle - Type == 'V' set spaces - Needed = 2. Else, set spaces - Needed = 1.
 - c) IF Process Emergency Rule == 'Y' Automatically declare transaction as valid.
 - d) IF profile exhibits non-permitted combinations, output Error warning message, increment

Roll no: 26k-0006

Total-Rejected and skip to next iteration.

e) Check Zone Space Availability

- If User = 'F': Check Zone A space. If $\text{ZoneA_Occ} + \text{Spaces_Needed} \leq \text{ZoneA_Max}$, allocate to zone A. Else if vehicle is Van, try Zone C if space exist. Else reject the vehicle.
- If User = 'S' with Permit 'Y' or a student Bike. Check Zone B space. If space allows, allocate to Zone B. Else reject the vehicle.
- If User = 'G' with Permit 'Y' check Zone C space. If space allows, allocate to Zone C, else reject the vehicle.

f) Update zone occupancy: $\text{Zone}[x]\text{-Occ} = \text{Zone}[x]\text{-Occ} + \text{Spaces_Needed}$ and increment Total Accepted.
Update Total-Processed = Total-Processed + 1.

5) END LOOP

6) Output final occupancy distribution, total accepted vs rejected counts, vehicle sub-distributions, and determine which zone holds the highest active occupancy.

7) End

Roll no: 26k-0006

PAC:

Given Data	Processing Required	Output Requirements
- Zone A=20, - Zone B=40, - Zone C=15 - Total_Expected count. - Vehicle_Type (C,B,V), User_Category (F,S,G), Permit (Y,N), Emergency (Y,N)	- Evaluate capacity space sizing (Vans take 2, others take 1) - Branch validation condition. - Check emergency status. - Sum total transactions and calculate space metrics tracking parameters.	- Vehicles validation - Comprehensive analytics report across zones.

IPO chart:

Input	Processing	Output
- Total vehicles loop limit count. - Vehicle_Type (C/B/V) - User_Category (F/S/G) - Permit_Status (Y/N) - Emergency_Status (Y/N)	- Initialize System Resource Allocation Arrays. - Evaluate structural layout for space availability - Calculate allocation destination using nested loops. - Increment internal counter values per iteration step. - Compute remainder space availability.	- Invalid credential alert message - End-of-cycle parking status report panel.

Roll no: 26k - 0006

Pseudocode:

BEGIN

DECLARE ZoneA-Max = 20, ZoneB-Max = 40, ZoneC-Max = 15 AS INTEGER

DECLARE ZoneA-Occ = 0, ZoneB-Occ = 0, ZoneC-Occ = 0 AS INTEGER

DECLARE Total-Processed = 0 AS INTEGER, Total-Expected AS INTEGER.

DECLARE Total-Accepted AS INTEGER, Total-Rejected AS INTEGER = 0.

DECLARE Cars, Bikes, Vans AS INTEGER = 0

DECLARE V-Type, U-Cat, Permit, Emer AS CHARACTER

DECLARE Space-Required AS INTEGER

PRINT "Enter number of vehicles"

INPUT Total-Expected.

WHILE Total-Processed < Total-Expected DO

PRINT "Enter Vehicle-Type (C/B/V), User-Category (F/S/G), Permit (Y/N), Emergency (Y/N)"

INPUT V-Type, U-Cat, Permit, Emergency

IF V-Type = 'V' THEN

Space-Req = 2

Vans = Vans + 1

Roll no: 26K-0006

ELSE

Space-Req = 1

IF $V_Type == 'C'$ THEN Cars = Cars + 1 Else Bikes = Bikes + 1

ENDIF

ENDIF

IF Emergency == 'Y' THEN

IF U-Cat == 'F' AND (ZoneA-Occ + Space-Req $\leq$ ZoneA_Max)

THEN ZoneA-Occ = ZoneA-Occ + Space-Req

Total-Accepted = Total-Accepted + 1

ELSEIF U-cat == 'S' AND (ZoneB-Occ + Space-Req $\leq$ ZoneB_Max)

THEN ZoneB-Occ = ZoneB-Occ + Space-Req

Total-Accepted = Total-Accepted + 1

ELSEIF U-cat == 'G' AND (ZoneC-Occ + Space-Req $\leq$ ZoneC_Max)

THEN ZoneC-Occ = ZoneC-Occ + Space-Req

Total-Accepted = Total-Accepted + 1

ELSE

PRINT "Alert! All zones entirely full"

Total-Rejected = Total-Rejected + 1

ENDIF

ELSE

IF (U-Cat == 'F' AND Permit == 'Y') THEN

IF ZoneA-Occ + Space-Req $\leq$ ZoneA_Max THEN

ZoneA-Occ = ZoneA-Occ + space-Req

Total-Accepted = Total-Accepted + 1

Roll no: 26k-0006

ELSE

PRINT "Zone A Full"

Total-Rejected = Total-Rejected + 1

ENDIF

ELSEIF (U-Cat == 'S' AND Permit == 'Y') OR (U-Cat == 'S'
AND (V-Type == 'B')) THEN

IF ZoneB-Occ + Space-Req <= ZoneB-Max THEN

ZoneB-Occ = ZoneB-Occ + Space-Req

Total Accepted = Total-Accepted + 1

ELSE

PRINT "Zone B Full"

Total-Rejected = Total-Rejected + 1

ENDIF

ELSEIF (U-Cat == 'G' AND Permit == 'Y') THEN

IF ZoneC-Occ + Space-Req <= ZoneC-Max THEN

ZoneC-Occ = ZoneC-Occ + Space-Req

Total-Accepted = Total-Accepted + 1

ELSE

PRINT "Zone C Full"

Total-Rejected = Total-Rejected + 1

ENDIF

ELSE

PRINT "Invalid Credentials, Entry Denied"

Total-Rejected = Total-Rejected + 1

ENDIF

ENDIF

Roll no: 26K-0006

Total-Processed = Total-Processed + 1

ENDWHILE

PRINT "... Final System Metrics Summary ... "

PRINT "Total-Processed: " Total-Processed, "

Accepted: " Total-Accepted, " Rejected: "

Total-Rejected

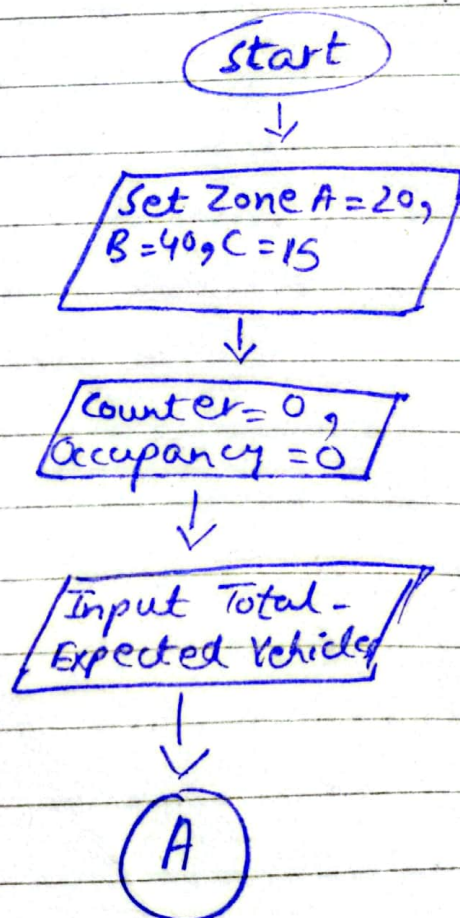
PRINT "FINAL Occupancies -> Zone A: " Zone A-Occ,

"/20, Zone B: " Zone B-Occ, "/40, Zone C: "

Zone C-Occ, "/15"

END

Flowchart:



Roll no: 26K-0006

